FULL-STACK PROTOTYPE · EXACT LOOKUP · LANGGRAPH

FU Berlin CS Consultant

A study consultant for the **FU Berlin Master Informatik** programme. It answers programme questions from local resources and validates proposed study plans against the deterministic 2014 Master Informatik rules.

Next.js · React · TypeScript | FastAPI · Python | LangGraph | Exact Course Lookup | Docker Compose

1 What it does

A backend-first study consultant with a standalone Next.js frontend. The LLM parses and explains; deterministic Python rules decide whether a study plan is valid.

Programme Q&A

Answers rule questions from the structured rule catalogue and course-offering questions from exact JSON buckets.

Study-Plan Check

Parses a proposed plan, validates LP totals and area rules, and explains issues.

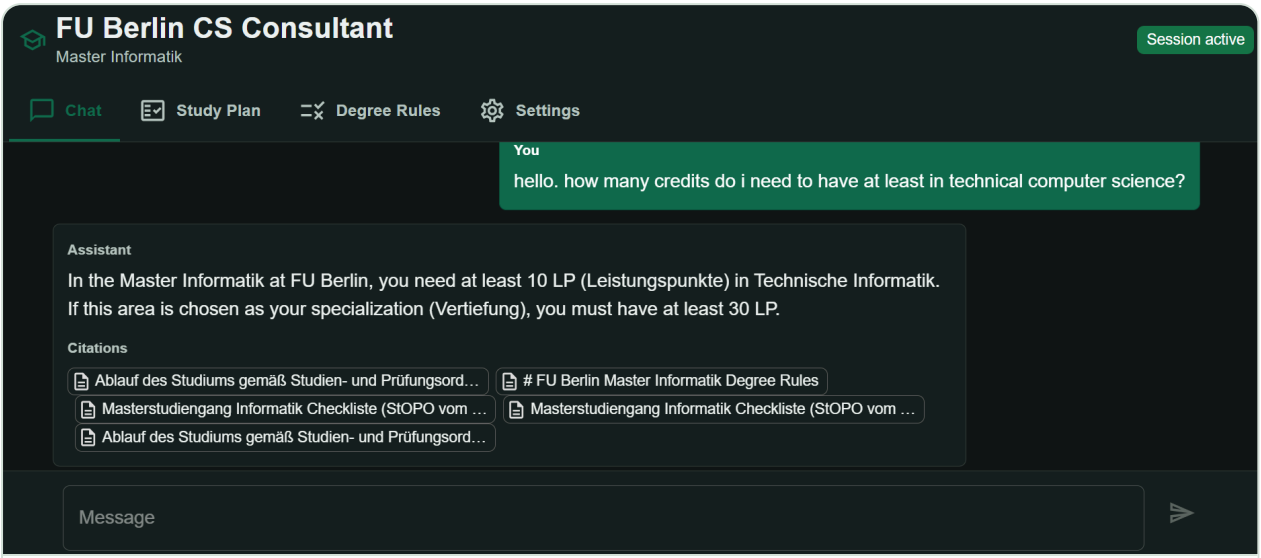
Degree Rules

Structured, read-only catalogue of programme rules served via a dedicated API.

> Advisory only — official FU Berlin documents and the examination office remain authoritative for real study decisions.

2 Chat in action

A real session in the consultant frontend — question, grounded answer, citations, and the four-tab shell around it.



Consultant chat tab - grounded answer with citations from local course data.

3 Architecture

Two services orchestrated by Docker Compose.

SERVICE	STACK	ENDPOINT
Frontend	Next.js App Router · React · TypeScript · Material UI	`http://localhost:3000/consultant`
Backend	FastAPI · LangGraph · Pydantic	`http://localhost:5100`

Browser + Frontend

Next.js UI at `:3000` . The browser calls the backend through the host-facing API URL.

Chat · Study Plan · Degree Rules · Settings

↓

FastAPI Backend

Owns sessions, transcript upload, LangGraph routing, rule validation, and response composition.

LangGraph · PDF parser · Rule checker · API

↓

Local Sources

Course buckets, program-rule catalogue, deterministic degree rules, and in-memory session state.

LLM Provider

AcademicCloud, FU Ollama over SSH, or local Ollama depending on `LLM_PROVIDER` .

Important Docker detail: `NEXT_PUBLIC_API_BASE_URL` points to `http://localhost:5100` because frontend JavaScript runs in the user's browser, not inside the Docker network.

Backend API

```
GET  /health
GET  /api/program-rules
POST /api/sessions
POST /api/sessions/{session_id}/message
POST /api/sessions/{session_id}/transcript
```

4 Agent Flow

A LangGraph state machine routes every user message to the right pipeline. Off-topic messages short-circuit. Study questions go through exact course lookup when needed; plan checks skip retrieval and go through deterministic validation.

START → ScopeClassifier

OFF_TOPIC → OfftopicReply → END

STUDY_QUESTION → CourseKeySelector → CourseLookup → AnswerComposer → END

PLAN_CHECK → StudyPlanParser → RuleChecker → AnswerComposer → END

```
LLM parses  →  Python validates  →  LLM explains
```

CourseKeySelector may return zero, one, or many lookup buckets such as `sose26/technical/swp`. CourseLookup then returns the complete selected buckets from local JSON.

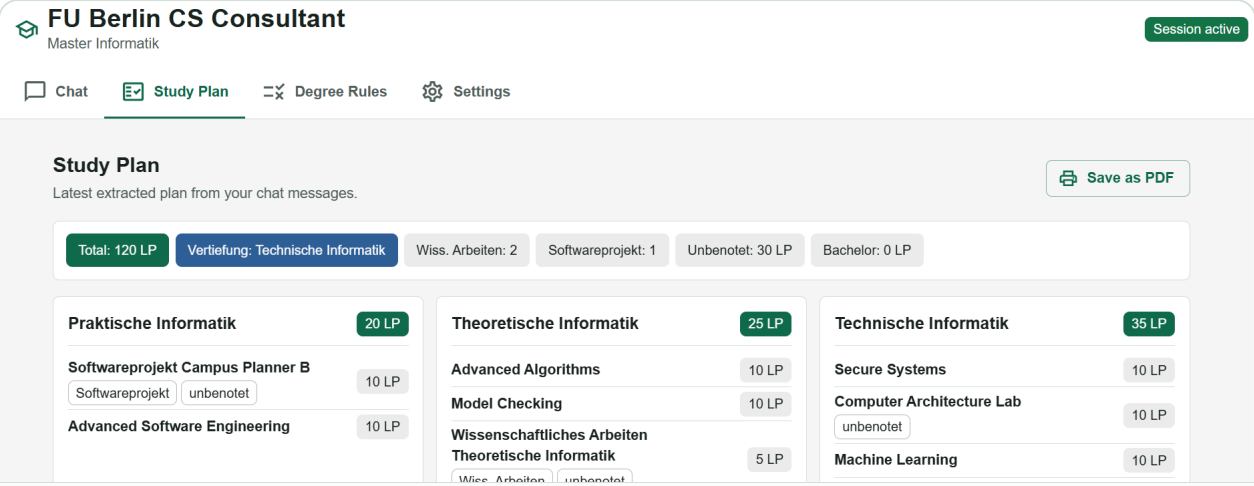
5 Study-Plan Validation

The rule checker is plain Python — auditable, deterministic, and never delegated to the LLM. It currently covers:

- 90 LP module area before thesis
- 30 LP Masterarbeit
- 70–80 LP Informatik area
- 10–20 LP Anwendungsbereich
- Specialization minimums
- Wissenschaftliches Arbeiten min/max
- Softwareprojekt min/max
- Wahlbereich seminar/project caveat
- 25–30 LP ungraded modules
- Max 15 LP Bachelor modules
- Duplicate module detection

6 Study Plan in action

The Study Plan tab renders the latest extracted plan with LP totals, the chosen Vertiefung, Wahlbereich data, and the per-area module breakdown.



Study Plan tab — extracted modules grouped by area with LP totals and Wahlbereich tags.

A **Save as PDF** button in the top-right exports the current study plan to a print-ready PDF for sharing with advisors or saving alongside official paperwork.

7 LLM Providers

Three interchangeable providers, selected via `LLM_PROVIDER`.

AcademicCloud

Hosted API, defaulting to `qwen3-235b-a22b`.

FU Ollama

FU `cuda01` reached automatically through an SSH tunnel.

Local Ollama

Ollama running on the host machine, reached via `host.docker.internal`.

8 Frontend

Next.js App Router with Material UI. The consultant app — not a marketing page — is the first screen. Four tabs cover the full workflow.

Chat

Session creation, message send, assistant response, citations and plan-check summaries.

Study Plan

Extracted modules, LP totals, Wahlbereich data and validation issues per student.

Degree Rules

Read-only rendering of `GET /api/program-rules`.

Settings

Dark mode plus developer tools — session ID, reset, backend health, API base URL.

9 Getting Started

Two commands to a running stack.

```
# 1. Configure the provider you want
cp .env.example .env

# 2. Build and start everything
docker compose up --build
```

Then open

Consultant — http://localhost:3000/consultant

API docs — http://localhost:5100/docs

Health — http://localhost:5100/health

10 Tech Stack

Next.js 14 · React 18 · TypeScript · Material UI · FastAPI · Pydantic · LangGraph · Ollama · Docker Compose · pytest

FU Berlin CS Consultant · prototype project · see agent-flow.html for the detailed routing diagram.